

Morphological boundary glottals in A'ingae: A new argument for [δ]*

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1. Introduction

This paper describes and analyzes patterns of allomorphy observed in the information structure (IS) domain of A'ingae (or Cofán, ISO 639-3: con). A'ingae has four IS suffixes: the new topic *-(?)ta* NEW,¹ contrastive topic *-(?)ja* CNTR, exclusive focus *-(?)yi* EXCL, and additive focus *-?khe* ADD. The first three of these markers show a regular alternation between plain (i. e. non-preglottalized; *-ta* NEW, *-ja* CNTR, *-yi* EXCL) and preglottalized (*-ʔta* NEW, *-ʔja* CNTR, *-ʔyi* EXCL) forms, conditioned by the syntactic category of the base of attachment.

When IS morphemes are attached to phrases of most syntactic categories, including—for example—noun phrases, overtly subordinated clauses, and adverbs, they are realized as plain. However, when they are attached to infinitive verbs, finite verbs, or non-verbal predicates in subordinate clauses, they are realized as preglottalized. This may give rise to striking minimal pairs, where non-verbal predicates (1b) may be distinguished from arguments (1a) solely by the presence of glottalization before the IS marker. Here, this is

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¹The following glossing abbreviations have been used: 1 = first person, 2 = second person, 3 = third person, ACC = accusative, ACC2 = accusative 2, ADD = additive focus, ADJ = adjectivizer, AIMP = attenuated imperative, ANIM = animate, APPR = apprehensional, ASSR = assertive, CAUS = causative, CNTR = contrastive topic, DAT = dative, DIST = distal, DS = different subject, EN = event nominalizer, EXCL = exclusive focus, FLAT = flat, FRST = frustrative, IF = conditional, IMP = imperative, INF = infinitive, INGR = ingressive, IPFV = imperfective, IRR = irrealis, ITER = iterative, LOC = locative, NEG = negative, NEW = new topic, PASS = passive, PERM = permissive, PL = plural, PLA = pluractional, PLS = plural subject, PROH = prohibitive, PROX = proximal, RCPR = reciprocal, SG = singular, SS = same subject, TENT = tentative, VDM = verbal diminutive, YNQ = polar interrogative.

illustrated with the new topic *-(?)ta* NEW (realized as *-(?)nda* NEW after nasal vowels).² The information structural markers are underlined throughout the paper.³

- (1) MINIMAL *-(?)TA*-PAIR ON A NOUN (NOMINAL ARGUMENT VS. PREDICATE + *-(?)TA* NEW)
 a. *tíse chán=da* =tsû jí-ya-mbi b. *tíse chán=?da* =tsû jí-ya-mbi
 3SG mother=NEW =3 come-IRR-NEG 3SG mother=NEW =3 come-IRR-NEG
 “His/her mother will not come.” “If she is a mother, she won’t come.”
 (2025-01-20(1)_m11)

The four markers under discussion (*-(?)ta* NEW, *-(?)ja* CNTR, *-(?)yi* EXCL, *-?khe* ADD) form a natural class—they all encode information structural meanings, attach to the same range of constituents, and always appear at the very end of a phrase. As such, the observed systematic alternation between the plain and preglottalized forms should not be understood as three independent instances of allomorphy, but rather attributed to an underlying morphosyntactic property shared by all the IS markers.

Concretely, I will propose that the glottal stop (*-(?)*) is a realization of a T-head conditioned by linear adjacency to IS morphology. The conditioning environment is formalized as a generalized *discourse* feature $[\delta]$ (Bossi and Diercks 2019, Mikkelsen 2015), which dominates all the more specific information structural features.⁴ In the previous literature, $[\delta]$ has been motivated by word-order facts; the current paper provides novel morphological evidence that discourse markers share a common morphosyntactic feature.

2. Language background

A’ingae (or Cofán ISO 639-3: con) is an endangered Amazonian isolate (AnderBois et al. 2019, Hammarström et al. 2020) spoken by ca. 1,500 Cofán people in Ecuador and Colombia. A’ingae syllable structure is (C)V(V)(?)—onsets are optional, nuclei are maximally diphthongal, and glottal stops are the only licit coda (Dąbkowski 2024b).⁵

A’ingae is highly agglutinating, exclusively suffixing, and encliticizing. There are many lexically contrastive morphemes that constitute plain–preglottalized minimal pairs, e. g. the flat classifier *-je* FLAT vs. imperfective *-?je* IPFV. Only the discourse markers *-(?)ta* NEW, *-(?)ja* CNTR, and *-(?)yi* EXCL participate in the specific alternation described in this paper.

The data presented in this paper comes from published materials written originally in A’ingae and fieldwork elicitation conducted by the author. Elicitation tasks included translation and grammaticality judgments. All the data drawn from previous publications are cited as such. All the fieldwork data has been deposited in the California Language Archive (CLA) as Dąbkowski (2020) and cited with a YYYY-MM-DD(N)_ccc identifier.

²The morphemes *-(?)ta* NEW, *-(?)ja* CNTR, and *-(?)yi* EXCL surface as *-(?)nda* NEW, *-(?)jan* CNTR, and *-(?)ñi* EXCL after nasal vowels due to nasal spreading (Dąbkowski 2024b, Sanker and AnderBois 2024, Bennett et al. 2024). The oral-nasal alternation is orthogonal to the plain–glottal alternation under scrutiny.

³Depending on the base of attachment, I represent functional morphemes as affixes or clitics. This does not correlate with the plain–glottal alternation of interest. For further discussion, see Dąbkowski (2025).

⁴Notwithstanding, I represent the glottal stop as part of the following IS morpheme in line with general glossing conventions adopted in the literature on A’ingae.

⁵I use the practical orthography with two deviations: glottal stops are represented with the IPA symbol *ʔ*, not apostrophe (*'*), and stress is marked with the acute accent (*´*). For more on A’ingae orthography, see Dąbkowski (2024b), Repetti Ludlow et al. (2019), Fischer and Hengeveld (2023).

3. Description

A'ingae distinguishes two types of overtly marked topics. The new topic *-(ʔ)ta* NEW indicates that the topic was not previously present in the discourse (2a). The contrastive topic *-(ʔ)ja* CNTR is used in the presence of alternative topics in the discourse (2b). The contributions of the topic markers are often not reflected in translations to Spanish or English, and most sentences are also accepted as grammatical without them.

- (2) TWO TOPIC MARKERS: *-(ʔ)TA* NEW AND *-(ʔ)JA* CNTR
- | | |
|---|---|
| <p>a. <i>yáya=ta</i> =<i>tsû tsámpì=ni já</i>
 dad=NEW =3 forest=LOC go
 “Dad went hunting.”
 (2024-06-10(1)_m11)</p> | <p>b. <i>áʔtse=ja</i> <i>tsáʔu=nga</i> =<i>tsû káʔni</i>
 h.bird=CNTR house=DAT =3 enter
 “A hummingbird entered the house.”
 (2024-04-03(1)_sia)</p> |
|---|---|

Focus indicates that in addition to the entity denoted by the marked phrase, there are others that are also relevant to the interpretation of the sentence (Krifka 2008:247). A'ingae has two focus morphemes. The exclusive focus *-(ʔ)yi* EXCL indicates a proposition holds of the marked entity to the exclusion of the alternatives; it is often translated as “only,” “just,” “very (same),” or with cleft constructions (3a). The additive focus *-ʔkhe* ADD indicates that the proposition holds of the marked entity in addition to alternatives; it is often translated as “too,” “as well” or “even” (3b). (Since the additive focus *-ʔkhe* ADD does not show the plain-glottal allomorphy of interest, I will not present further examples with it. However, its non-alternating behavior is explained in section 4.)

- (3) TWO FOCUS MARKERS: *-(ʔ)YI* EXCL AND *-ʔKHE* ADD
- | | |
|---|--|
| <p>a. <i>ñá=ñi</i> <i>já-ye inʔjan</i>
 1SG=EXCL go-INF want
 “Only I want to go.”
 (2023-12-11(2)_rgq)</p> | <p>b. <i>ñá=ʔkhe</i> <i>jénʔtshi-ye atésû</i>
 1SG=ADD sneeze-INF know
 “I, too, usually sneeze.”
 (2023-08-29(3)_eol)</p> |
|---|--|

3.1 Non-predicates

The A'ingae IS morphemes are most frequently observed on various arguments and adjuncts, including e. g. bare nouns (2), pronouns (3), question words (4a), adjectives, and adverbs (4b). In all these cases, they are realized as plain (i. e. not preglottalized).

- (4) PLAIN REALIZATION ON NON-PREDICATES
- | | |
|---|--|
| <p>a. <i>junguésû=yi</i> =<i>tsû náʔen=ni kánse?</i>
 what=EXCL =3 river=LOC live
 “What is it that lives in the river?”
 (2024-06-10(1)_m11)</p> | <p>b. <i>tayúpi-ta</i> =<i>tsû Erisión tsáiʔmbi-ʔtshi teteté=ndekhû=ve</i> <i>fithi~ʔthi</i>
 long ago-NEW =3 Erisión many-ADJ Waorani=PL.ANIM=ACC2 kill~PLA
 “Once upon a time, Erisión killed many Waoranis.”
 (Blaser and Chica Umenda 2008:152; 2024-06-07(1)_m11)</p> |
|---|--|

3.2 Predicates

In addition to arguments and adjuncts, the IS morphemes may appear on subordinate predicates, both verbal and non-verbal. A’ingae predicates can vary greatly in morphological complexity (Dąbkowski 2021, 2024c). On one end, the head of a finite TP may consist of a bare root. On the other end, a plethora of grammatical categories can be expressed on a verb by means of suffixation (5).

(5) VERBAL PREDICATE STRUCTURE

[[[[[*kufi* -án]_{VceP} -ʔje -ngi]_{AspP} -ʔfa -mbi]_{TP} -ʔni]_{CP} -nda]_{ΔP}
 play -CAUS -IPFV -PROX -PLS -NEG -IF.DS -NEW

“now_{NEW}, if_{IF} (they_{PLS}) do not_{NEG} come_{PROX} to be_{IPFV} making_{CAUS} play, (someone else_{DS}) ...” (2024-06-18(1)_m11)

The suffixes can be grouped into five major functional projections. The A’ingae verbal template is given in Table 1. The verbal root is at the bottom, and the subsequent morphosyntactic slots appear above it, mimicking the orientation of a syntax tree.

A’ingae lacks a copula verb; as such, the TP, CP, and ΔD morphemes attach directly to non-verbal predicates. IS morphemes behave in similar ways on verbal and non-verbal pred-

INFO STRUCTURAL SUFFIXES (ΔP)

(xii) TOPIC: -ʔta NEW, -ʔja CNTR

(xi) ADDITIVITY: -ʔkhe ADD

(x) EXCLUSIVITY: -ʔyi EXCL

CLAUSE-LEVEL SUFFIXES (CP)

(ix) CLAUSE TYPE

MATRIX: -ja IMP, -kha AIMP, -ʔse PERM, -jama PROH, -ʔya ASSR

COSUBORDINATE: -pa SS, -si DS

SUBORDINATE: -saʔne APPR, -khen TENT, -ʔni IF.DS, -ʔma FRST

SITUATION-LEVEL SUFFIXES (TP)

(viii) POLARITY: -mbi NEG

(vii) REALITY/FINITENESS: -ya IRR, -ye INF

(vi) SUBJECT NUMBER: -ʔfa PLS

VERBAL INFLECTIONAL SUFFIXES (AspP)

(v) ASSOC MOTION: -ʔngi PROX, -ʔnga DIST

(iv) ASPECT: -ʔje IPFV, -ji INGR, -kha VDM, -ʔñakha ITER

VOICE SUFFIXES (VceP)

(iii) PASSIVE: -ye PASS

(ii) RECIPROCAL: -khu RCPR

(i) CAUSATIVE: -ñā/-an/-en CAUS

VERBAL ROOT (√P)

(o) VERBAL ROOT: √

Table 1: Morphological template of the A’ingae verb (building on Dąbkowski 2024c)

icates. As such, in the rest of the paper, I group and discuss them together. A'ingae subordinate clauses can be inflected for features such as subject plurality (-ʔfa PLS), reality status (-ya IRR), polarity (-mbi NEG), and finiteness (-ye INF). This suggests that subordinate clauses consist of at least the TP layer (and also may have an overt or phonologically null CP layer).

3.2.1 Infinitive predicates

First, I considered the various uses of infinitive predicates. Infinitive clauses may, for example, function as subjects of stative predicates (6a), arguments selected by verbs such as the habitual auxiliary *atesû* 'know' (6b), or rationale clauses (6c).

(6) PREGLOTTALIZED REALIZATION ON INFINITIVES WITH -YE INF

- a. *án-ñe-ʔnda* =*tsû injéng-e-ʔchu* b. *atesû* =*ngi guáʔthi-an-ñe-ʔjan*
eat-INF-NEW =3 needed-EN know =1 boil-CAUS-INF-CNTR
"Eating is important."
(2024-05-27(2)_sia) "I (habitually) boil."
(2024-06-11(2)_m11)
- c. *júsû afa-khú-ye-ʔyi-ta* =*ngi áʔingae=ma atésû-ʔje*
only speak-RCPR-INF-EXCL-NEW =1 A'ingae=ACC learn-IPFV
"I am studying A'ingae only so that I can argue (with people)."
(2025-01-20(1)_m11)

All of these uses are compatible with IS markers. When appearing on an infinitive verb, they are always realized as preglottalized (6a, 6b). If multiple IS markers appear on a predicate (such as an infinitive), a glottal stop is observed only before the first one (6c).

3.2.2 Finite predicates

Now, I will look at finite predicates with IS morphemes. A'ingae finite clauses may be (co)subordinated e. g. with the same-subject marker *-pa* SS (7a), different-subject marker *-si* DS (7b), apprehensional *-saʔne* APPR (7c), tentative *-khen* TENT (7d), or different-subject conditional *-ʔni* IF.DS (7e), which may (but need not) be followed by IS markers. (The same-subject conditional—to be discussed momentarily—will be analyzed as phonologically null). After the overt subordinators, IS markers are realized as non-preglottalized.

(7) PLAIN REALIZATION ON OVERTLY SUBORDINATED CLAUSES

- a. *amphí-pa-ta* =*ti=ki tsífu=ja báthi-ʔchu-mbi?*
fall-SS-NEW =YNQ=2 neck=CNTR dislocate-EN-NEG
"Did you injure your neck by falling?"
(Borman et al. 1991:60; 2024-06-07(1)_m11)
- b. *kéʔi atesû-ʔfa-mbi-si-ja* *kéʔi=nga teváen-mbi* =*ngi*
2PL know-PLS-NEG-DS-CNTR 2PL=DAT write-NEG =1
"I do not write to you because you do not know (the truth)."
(1 John 2:21; 2024-06-10(1)_m11)

- c. *tiseʔpa ña yayá-ndekhú-ʔfa-saʔne-ñi* =ngi dyúju
 3PL 1SG dad-PL.ANIM-PLS-APPR-EXCL =1 be afraid
 “I fear only that they are my parents.” (2024-06-11(1)_m11)
- d. *athe-yé-khen-nda* =ngi tsún-ʔjen e. *thési=ma áthe-ʔfa-ʔni-ñi* =tsû búthu
 see-PASS-TENT-NEW =1 do-IPFV jaguar=ACC see-PLS-IF.DS-EXCL =3 run
 “I am trying to be seen.” “As soon as they saw a jaguar, (sb
 (2024-06-11(2)_m11) else) ran.” (2024-06-18(1)_m11)

A different pattern is seen with same-subject conditional antecedents, as they do not receive any dedicated overt marking. Rather, same-subject antecedents are marked only with an IS morpheme. When introducing a same-subject conditional antecedent, the first IS marker is always preglottalized (8a, 8b). The following IS markers, if present, surface without an additional glottal stop (8c). A same-subject antecedent is always introduced by at least one IS morpheme.

(8) PREGLOTTALIZED REALIZATION ON SAME-SUBJECT CONDITIONALS

- a. *ña yáya=ʔta* =tsû afé-ya fae regalo=ve ña=nga
 1SG dad=NEW =3 give-IRR one gift=ACC2 1SG=DAT
 “If he’s my dad, he’ll give me a gift.” (2024-06-10(1)_m11)
- b. *afa-khú-ʔja* sumbú-ya =ngi
 speak-RCPR-CNTR leave-IRR =1
 “If I argue, I will leave.” (2024-06-11(2)_m11)
- c. *thési=ma áfase-ʔyi-ja* búthu-ya =ngi
 jaguar=ACC criticize-EXCL-CNTR run-IRR =1
 “Only if I criticize a jaguar, I will run.” (2024-06-06(2)_m11)

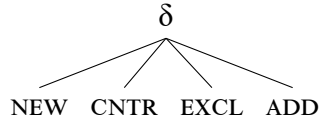
In an interim summary, in most contexts, the IS markers are not preceded by a glottal stop. The glottal stop appears if an IS marker attaches directly to a predicate, including (i) infinitival clauses and (ii) morphologically unmarked same-subject conditional antecedents.

4. Analysis

I propose that the four IS morphemes (-*ta* NEW, -*ja* CNTR, -*yi* EXCL, -*ʔkhe* ADD) are the exponents of four discourse features: [NEW], [CNTR], [EXCL], and [ADD]. The discourse features are organized hierarchically, dominated by a superordinate *discourse* feature [δ] (9).

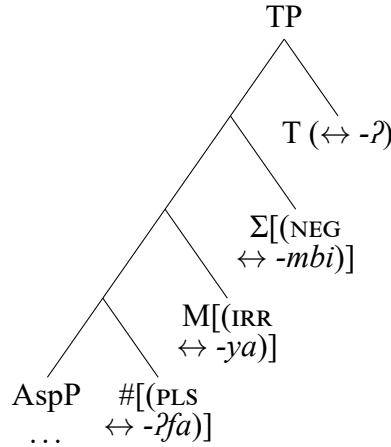
The hierarchy in (9) draws on Mikkelsen (2015) and Bossi and Diercks (2019)’s accounts of word order in, respectively, Danish and Kipsigis. In Kipsigis (Nilo-Saharan, Kenya), the immediately post-verbal position is occupied by a discourse-prominent item (Bossi and Diercks 2019). To account for this pattern, Bossi and Diercks (2019) introduce “an underspecified [δ] (discourse) feature that can be satisfied by phrases of any information structure designation” (p. 30). My proposal adopts their feature hierarchy directly.

(9) DISCOURSE
FEATURE HIERARCHY

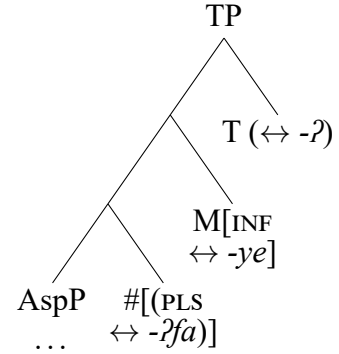


(10) TP STRUCTURE

a. FINITE TPs



b. INFINITIVE TPs



I assume a decompositional approach to T, where overt TAM suffixes are realized as heads below T. The specific hierarchies of projections are given in (10).⁶Crucially, I propose that the glottal stop $-ʔ$ is a contextual realization of T° . Specifically, T° is realized as a glottal stop $-ʔ$ when linearly adjacent to the discourse feature $[\delta]$. The critical vocabulary item (VI) is given in (11a). Since $-ta$ NEW, $-ja$ CNTR, $-yi$ EXCL, and $-ʔkhe$ ADD all inherit from $[\delta]$, they all satisfy this environment condition. Otherwise, T° is realized as phonologically null (11b). This derives the observed distribution of the IS-conditioned $-ʔ$. Some of the relevant vocabulary items are given in (11-13).

(11) FEATURES $\preceq$ TP

- a. T ↔ $-ʔ$ / $_ \delta$
- b. T ↔ $-\emptyset$ / elsw.
- c. PLS ↔ $-ʔfa$
- d. IRR ↔ $-ya$
- e. INF ↔ $-ye$

(12) FEATURES $\preceq$ CP

- a. IF, SS ↔ $-\emptyset$
- b. SS ↔ $-pa$
- c. IF, DS ↔ $-ʔni$
- d. DS ↔ $-si$
- e. APPR ↔ $-saʔne$

(13) FEATURES $\preceq$ ΔP

- a. NEW ↔ $-ta$
- b. CNTR ↔ $-ja$
- c. EXCL ↔ $-yi$
- d. ADD ↔ $-ʔkhe$

Now, I demonstrate how the analysis accounts for the data. When an IS marker attaches to most syntactic categories, such as DPs (14a), adjectives, or adverbs, there is no T° adjacent to a discourse-marked morpheme. As such, $-ʔ$ is not realized. When an IS marker attaches immediately to an infinitive TP, such as a complement of a raising predicate (14b), the adjacent T-head is realized as $-ʔ$ (11a). The base of attachment is bracketed [].

(14) $-ʔ$ NOT REALIZED ON A DP VS. $-ʔ$ REALIZED ON A RAISING PREDICATE COMPLEMENT

- a. $[yáya]_{DP=ta} = tsû tsámpi = ni \quad já$
dad=NEW =3 forest=LOC go
“Dad went hunting.”
- b. $[ñúʔfa-ye-ʔ]_{TP=ta} = ngi atésû$
rest-INF-T-NEW =1 know
“I (habitually) rest.”

(2024-06-10(1)_m11)

(2024-10-08(1)_m11)

When an IS marker attaches to CP with overt subordinating C°, there is overt material intervening between T° and the IS marker (15a). Now, since environments conditioning allomorphy are strictly local (Embick 2010, 2015), -ʔ is not realized (11b).⁷

(15) OVERTLY-SUBORDINATED CLAUSE: -ʔ NOT REALIZED

- a. [já-ya-Ø-pa]_{CP-yi} =ngi ina-ʔjen
go-IRR-T-SS-EXCL =1 cry-IPFV
“I’m crying only because I will leave.” (2024-10-08(1)_m11)

However, when an IS marker attaches to a CP with a null C°, such as a complement of a control predicate (16a), there is no overt morphology intervening between T° and the IS marker. Since phonologically unrealized material is ignored for purposes of satisfying allomorphy environments (*pruning* in Embick 2010, 2015), T° is realized as -ʔ.

I propose that same-subject conditional antecedents are likewise introduced by a head that is phonologically unrealized (12a).⁸ Since, again, null material is ignored for the purposes of allomorphy (Embick 2010, 2015), T° is realized as -ʔ (16b).

(16) CONTROL PREDICATE COMPLEMENT OR SAME-SUBJECT ANTECEDENT: -ʔ REALIZED

- a. [kéʔi án-ʔfa-ye-ʔ-Ø]_{CP-ja} séʔpi =ngi b. [afa-khú-ʔ-Ø]_{CP-ja} sumbú-ya =ngi
2PL eat-PL-INF-T-C-CNTR forbid =1 speak-RCPR-T-IF.SS-CNTR leave-IRR =1
“I prohibit y’all from eating.” “If I argue, I will leave.”
(2024-10-08(1)_m11) (2024-06-11(2)_m11)

This derives the fact that preglottalization is realized on same-subject antecedents, but not on other types of subordinate clauses—only the same-subject antecedents are introduced by a phonologically unrealized feature bundle.⁹

5. Discussion and conclusions

The above analysis draws on the notion of a superordinate discourse feature [δ] (Bossi and Diercks 2019, Mikkelsen 2015), which has previously been motivated by word-order facts. A’ingae provides novel morphological evidence for [δ].

⁶I assume that every clause—by definition—contains the TP layer (Shlonsky 1997:3). Additionally, I assume that TP dominates the projections that realize negation (Laka Mugarza 1990, Pollock 1989), irrealis mood (Cinque 1999), and finiteness (Wurmbrand 1998). Finally, I assume that the hierarchical order of the number (#P), mood (MP), and polarity (ΣP) projections mirrors the linear order of suffixes they introduce (-ʔfa PLS, -ya IRR, and -mbi NEG). While T° is often silent, its presence correlates in A’ingae with specific temporal interpretations. For example, while uninflected stative and non-verbal TPs are interpreted as present, uninflected eventive verbs are interpreted as realis, perfective, and past. For more, see Dąbkowski (2024a).

⁷The assumption that infinitival morphology is TP-internal, while subordinating morphemes are C-heads, is accepted cross-linguistically (Adger 2003), and corroborated for A’ingae by Dąbkowski (2024c, 2022).

⁸Cross-linguistically, conditionals and topics are marked in similar, often identical, ways (Haiman 1978). In A’ingae, the IS morphemes appear optionally (though frequently) on different-subject antecedents (7e), and obligatorily on same-subject antecedents (8). Nonetheless, conditionality is expressed with dedicated morphosyntactic features (12a, 12c).

⁹The additive focus -ʔkhe ADD is always realized as preglottalized. I propose that -ʔkhe ADD is underlyingly preglottalized (13d). I assume that it participates in the same triggering of T-allomorphy as the other IS morphemes, but two contiguous glottal stops are later phonologically reduced to one.

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Descriptively, the proposed analysis makes a non-obvious claim: Even though same-subject (SS) conditionals are most often introduced by the preglottalized *-ʔta* NEW and *-ʔja* CNTR, neither morpheme is analyzed as contributing the conditional meaning. Rather, *-ʔ* is proposed to be a syntactically conditioned spell-out of a functional head, the IS marker contributes its regular discourse meaning, and the SS conditional morpheme is phonologically silent (16b). This shows that language description and theoretical analysis are not two separate endeavors, and that they must mutually inform each other.

References

- Adger, David. 2003. *Core syntax: A Minimalist approach*. Oxford Core Linguistics. Oxford University Press.
- AnderBois, Scott, Nicholas Emlen, Hugo Lucitante, Chelsea Sanker, and Wilson de Lima Silva. 2019. Investigating linguistic links between the Amazon and the Andes: The case of A'ingae. Paper presented at the 9th Conference on Indigenous Languages of Latin America, University of Texas at Austin.
- Bennett, Ryan, Shen Aguinda, Hugo Lucitante, and Scott AnderBois. 2024. Anticipatory nasalization in A'ingae. Talk presented at the Phonetics and Phonology Lunch at the University of California, Santa Cruz.
- Blaser, Magdalena, and María Enma Chica Umenda. 2008. *A'indeccu canque'sune condase'cho. Mitos del pueblo cofán*. Centro Cultural de Investigaciones Indígenas Padre Ramón López.
- Borman, Roberta “Bobbie”, Verla Cooper, and Emeregildo Criollo. 1991. *El cofán y el médico: Vocabulario médico bilingüe cofán–castellano*. Quito, Ecuador: Instituto Lingüístico de Verano (Summer Institute of Linguistics).
- Bossi, Madeline, and Michael Diercks. 2019. V1 in Kipsigis: Head movement and discourse-based scrambling. *Glossa: A Journal of General Linguistics* 4.
- Cinque, Guglielmo. 1999. *Adverbs and functional heads: A cross-linguistic perspective*. Oxford Studies in Comparative Syntax. Oxford University Press.
- Dąbkowski, Maksymilian. 2020. A'ingae Field Materials. California Language Archive, Survey of California and Other Indian Languages, University of California, Berkeley. 2020–19. <http://dx.doi.org/doi:10.7297/X2HH6HKG>.
- Dąbkowski, Maksymilian. 2021. Dominance is non-representational: Evidence from A'ingae verbal stress. *Phonology* 38:611–650. <https://www.doi.org/10.1017/S0952675721000348>.
- Dąbkowski, Maksymilian. 2022. A'ingae pied-piping: A Q-based analysis. Paper presented at the 4th Symposium on Amazonian Languages, University of California, Berkeley. <https://lingbuzz.net/lingbuzz/008650>.
- Dąbkowski, Maksymilian. 2024a. Phonology grants no asylum: The inescapability of syntax in A'ingae dominance blocking. Paper presented at the workshop *Phonological domains and what conditions them*, University of California, Berkeley. <https://lingbuzz.net/lingbuzz/008770>.

- Dąbkowski, Maksymilian. 2024b. The phonology of A'ingae. *Language and Linguistics Compass* 18:e12512. <http://dx.doi.org/10.1111/lnc3.12512>.
- Dąbkowski, Maksymilian. 2024c. Two grammars of A'ingae glottalization: A case for Cophonologies by Phase. *Natural Language and Linguistic Theory* 42:437–491. <https://doi.org/10.1007/s11049-023-09574-5>.
- Dąbkowski, Maksymilian. 2025. Metrical stress and glottal stops in A'ingae: A study of cyclicity and dominance at the interface of phonology and morphology. Doctoral dissertation, University of California, Berkeley. URL <https://lingbuzz.net/lingbuzz/009397>.
- Embick, David. 2010. *Localism versus Globalism in Morphology and Phonology*. Number 60 in Linguistic Inquiry Monograph. Cambridge, MA: MIT Press.
- Embick, David. 2015. *The Morpheme: A Theoretical Introduction*. Number 31 in Interface Explorations [IE]. Boston and Berlin: De Gruyter Mouton.
- Fischer, Rafael, and Kees Hengeveld. 2023. A'ingae (Cofán/Kofán). In *Language Isolates I: Aikanã to Kandozi-Shapra*, ed. by Patience Epps and Lev Michael, volume 1 of *Handbooks of Linguistics and Communication Science (HSK)*, 65–124. Berlin: De Gruyter Mouton. <https://doi.org/10.1515/9783110419405-002>.
- Haiman, John. 1978. Conditionals are topics. *Language* 54:564–589. <https://doi.org/10.2307/412787>.
- Hammarström, Harald, Robert Forkel, Martin Haspelmath, and Sebastian Bank, ed. by. 2020. *Glottolog 4.3 - Cofán*. Jena: Max Planck Institute for the Science of Human History. <https://glottolog.org/resource/languoid/id/cofa1242#refs>.
- Krifka, Manfred. 2008. Basic notions of information structure. *Acta Linguistica Hungarica* 55:243–276.
- Laka Mugarza, Miren Itziar. 1990. Negation in syntax: On the nature of functional categories and projections. Doctoral dissertation, Massachusetts Institute of Technology.
- Repetti Ludlow, Chiara, Haoru Zhang, Hugo Lucitante, Scott AnderBois, and Chelsea Sanker. 2019. A'ingae (Cofán). *Journal of the International Phonetic Association: Illustrations of the IPA* 1–14. <https://doi.org/10.1017/S0025100319000082>.
- Mikkelsen, Line. 2015. VP anaphora and verb-second order in Danish. *Journal of Linguistics* 51:595–643. <https://doi.org/10.1017/S0022226715000055>.
- Pollock, Jean-Yves. 1989. Verb movement, universal grammar, and the structure of IP. *Linguistic inquiry* 20:365–424. <http://www.jstor.org/stable/4178634>.
- Sanker, Chelsea, and Scott AnderBois. 2024. Reconstruction of nasality and other aspects of A'ingae phonology. *Cadernos de Etnolingüística* 11:e110101. <http://www.etnolinguitica.org/article:vol11n1>.
- Shlonsky, Ur. 1997. *Clause structure and word order in Hebrew and Arabic: An essay in comparative Semitic syntax*. Oxford University Press.
- Wurmbrand, Susanne. 1998. Infinitives. Doctoral dissertation, Massachusetts Institute of Technology. <http://hdl.handle.net/1721.1/9592>.